mechanics, string theory has revealed that the fabric of the cosmos may have many more dimensions than we perceive directly—dimensions that may be the key to resolving some of the universe's deepest mysteries. Moreover, the theory intimates that the familiar notions of space and time do not extend into the sub-Planckian realm, which suggests that space and time as we currently understand them may be mere approximations to more fundamental concepts that still await our discovery.

In the universe's initial moments, these features of the spacetime fabric that, today, can be accessed only mathematically, would have been manifest. Early on, when the three familiar spatial dimensions were also small, there would likely have been little or no distinction between what we now call the big and the curled-up dimensions of string theory. Their current size disparity would be due to cosmological evolution which, in a way that we don't yet understand, would have had to pick three of the spatial dimensions as special, and subject only them to the 14 billion years of expansion discussed in earlier chapters. Looking back in time even further, the entire observable universe would have shrunk into the sub-Planckian domain, so that what we've been referring to as the fuzzy patch (in Figure 10.6), we can now identify as the realm where familiar space and time have yet to emerge from the more fundamental entities—whatever they may be—that current research is struggling to comprehend.

Further progress in understanding the primordial universe, and hence in assessing the origin of space, time, and time's arrow, requires a significant honing of the theoretical tools we use to understand string theory—a goal that, not too long ago, seemed noble yet distant. As we'll now see, with the development of M-theory, progress has exceeded many of even the optimists' most optimistic predictions.